

Sayandip Halder

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EXPERIENCE

CoverPin

Software Engineer

May 2025 – Present

San Francisco Bay Area, Remote

- Designed and implemented a complete **Role-Based Access Control (RBAC)** backend system to enforce workspace-level resource restrictions; strengthened enterprise readiness, accelerated deal closures, and supported large-scale onboarding of customers with hundreds of entities and locations.
- Designed a **stack based data structure** that leverages **React Context** API to manage sequential modal components in a React app. It provides a mechanism to simplify complex, multi-step user interaction flows using modals.
- Led end-to-end development** of a scalable License Management module and a Locations module using Next.js, NestJS, and PostgreSQL, enabling customers to create business licenses, operational locations, manage license versions, and map them to entities; this directly contributed to onboarding 500+ new licenses and locations from new customers onto the platform.

Pepper Content

SDE1

June 2022 – April 2025

Mumbai, Remote

- Revamped the **creator onboarding journey**. First by reducing friction by minimizing steps and implementing structured data collection; improved user experience while standardizing backend data models. Then built a **KYC verification workflow** collecting bank, GST, and Aadhaar details to ensure regulatory compliance and prevent payouts to mismatched bank accounts, reducing operational and legal risk.
- Took serious ownership of and maintained the internal operations platform and the pepper creator platform; shipped new features, enforced best practices.

SIDE PROJECTS

IntelliDoc v1

Agentic RAG Platform

Python, FastAPI, LangGraph

- Built an **agentic RAG pipeline** that ingests documents, processes them through a multi-stage **LangGraph** agent workflow, and enables intelligent querying over long-form context.
- Implemented a **self-correcting CRAG loop** with retrieval evaluation and response refinement to improve answer faithfulness and retrieval quality.

RateLim

Distributed Rate Limiter

NestJS, Typescript, Redis

- Built a distributed **rate limiting system** for both API gateway and microservice layers using **NestJS** and **Redis**.
- Implemented a centralized **token bucket algorithm** with Redis as the source of truth to ensure consistent throttling across distributed services.

INTERNSHIPS

Google Summer of Code 2021

Student Developer

Summer 2021

Remote

- Contributed to **PyData/Sparse** under the aegis of the Python Software Foundation, working on a library positioned to become the de-facto sparse array implementation in the Python ecosystem, replacing `scipy.sparse`.
- Designed and implemented from scratch an **ndarray-compatible Python API**, enabling seamless integration with NumPy-like workflows.

EDUCATION

Jadavpur University

Bachelor's in Engineering

2018-2022

Kolkata, WB

SKILLS

Technologies : Typescript, JavaScript, Nestjs, Reactjs, Nextjs, Node, Express, HTML/CSS, SQL (PostgreSQL, MySQL), Python, FastAPI, LangChain, LangGraph, Java, Golang, MongoDB

Tools : Docker, Figma, Notion, Jira, Git, GitHub, Linear